Close to the cap: who plays the whole match across nine sports

Abstract

Governing bodies keep changing the cap, the number of changes a team may make. We read the rulebooks and match records of nine sports, over a million appearances. From the rulebook alone, a governing body can work out the smallest share of starters a new cap leaves on for the whole match. Teams came close to it under the caps they had. Under caps of ten or fewer, about half or more of the starters played the whole match, even where players could return. But where players may return freely or under a very large cap, almost every starter comes off. Where we compared positions under small caps, position mattered more than the cap. Where a replaced player cannot return, a team's changes and its starters' minutes must agree. This gives data providers a free check. In one large public dataset, it flags 8 to 27 per cent of teams' records. Each flagged record credits a starter taken off at minute 90 or later with the whole match. Where rules limit the pauses for changes, they force most of the bunching. A model of these pauses met all six predictions we wrote down before testing it on unseen matches.

Keywords: substitution rules; sports rules; match records; order statistics; underdispersion; data quality.

1 Introduction

Most team sports let a team replace players during a match. Most of them also limit how many, and we call this limit the *substitution limit*, or the *cap*. Governing bodies keep changing it. Football moved from three substitutions to five (The International Football Association Board, 2020, 2022), and rugby union lengthened its bench to eight replacements. Rugby league cut its cap to eight interchanges, and Australian rules cut its cap twice, to 75. In football, rugby union, Gaelic football, hurling and baseball, a replaced player cannot come back. We call these *one-way sports*, and Table 1 lists their few exceptions, such as replacements for blood injuries. In rugby league, Australian rules, ice hockey and basketball, a player may come back on. We call these four *re-entry sports*.

Each change of cap matters to four groups. A governing body needs to know how many starters a new cap leaves on for the whole match. A club needs to know which positions play the whole match, so that it can plan their load. Anyone who learns from match data needs to know whether the records say who really played the whole match (Pappalardo et al., 2019; Wang et al., 2024). And a match simulator needs to know how many changes teams make, and when (Titman et al., 2015; Maia et al., 2026). This paper answers these four questions for nine sports: football, rugby union, rugby league, Australian rules, Gaelic football, hurling, baseball, basketball and ice hockey.

A substitution limit works like a speed limit. Such a limit does not set how fast a driver goes, but it caps how far a car can travel in an hour. In the same way, a cap does not choose which players a coach takes off, but it caps how many starters a team can take off. Teams drive close to their cap. Under five substitutions, football's teams take off 39 per cent of their starters by substitution, and the cap allows 45.5

per cent.

Where a replaced player cannot return, simple counting links the starters who leave to the changes a team makes. Each change takes off either a starter or a player who came on earlier. So the starters who leave equal the team's changes, less the changes that take off an earlier substitute, plus the starters sent off. An equation that holds by counting alone, like this one, is called an *identity*. It gives governing bodies a formula. It also gives a free check on match records, because a record whose changes and minutes disagree is inconsistent. This matters, because rule changes for player safety have been judged on public match records (Beaudouin et al., 2019).

What this means in practice. The results answer the four questions.

- *For governing bodies.* From the cap alone, a governing body can compute the most starters a new cap lets a team take off. Teams come close to that maximum, because they use most of their cap and coaches rarely replace an earlier substitute. Under each cap of ten or fewer measured here, about half or more of the starters play the whole match, even where players may come back on. In rugby league most starters still play the whole match under its eight interchanges. In football the starters taken off and the changes made differ by less than 3 per cent of the team, in each month of the season.

- *For clubs.* In football and rugby league, who plays the whole match depends more on position than on the cap. Football's goalkeepers, rugby league's backs and baseball's starting batters finish far more often than outfield starters, forwards and starting pitchers. So their load is managed between matches, not by substitution.

- *For data providers.* A record's changes and its players' minutes must tell the same story. That simple check flags 8 to 27 per cent of

the teams' records in nine football leagues. In each flagged record, a starter taken off at minute 90 or later is credited with the whole match. The rugby league and Australian rules records pass a second check on average: their minutes sum to the players on the field times the length of the match.

- *For simulators.* We summarize the number of changes a team makes with a curve cut off at the cap. Its width is about one to two changes, and its centre moves with the limit. A team's changes come, on average, near where the same number of random moments in one stretch of the match would fall. A model of the stoppages, the pauses in which a team makes its changes, met all six pass marks written down before we read the matches kept back to test it.

What is new. Earlier work has studied substitutions mostly as a coaching decision. It has modelled the best moments to change players (Hirotsu & Wright, 2002, 2005; Hirotsu et al., 2024), timed the changes (Del Corral et al., 2008; Myers, 2012; Silva & Swartz, 2016; Hills et al., 2019; Sun et al., 2024) and related them to the score (Gómez et al., 2016; Wittkugel et al., 2022). It has measured how their use grew with each change of rule (Nevill et al., 2026; Dilger & Vischer, 2024), how often teams use their whole limit (Ribeiro et al., 2020; Meyer & Klatt, 2021; Roberts et al., 2024; World Rugby, 2020) and how often they make two changes at once (Xiao & Zhang, 2024). It has also compared rugby league's whole-match and rotated players (Black & Gabbett, 2014; LeagueEyeTest, 2021), and the limits of different sports (Mota et al., 2020). None of it asks what the rule itself caps. This paper studies a rule already in force, not a proposal for a new one (Wright, 2014).

We know of no earlier work, apart from the preprint described below, that does any of five things this paper does. It compares who

plays the whole match position by position across sports. It checks match records against the identity. It counts how often a substitute is replaced in turn. It models how many changes teams make and how they group them into stoppages. And it tests that model on matches the model had not seen. Ricou (2026) first reported, sport by sport, how many starters play the whole match and how closely the starters taken off track the substitutions made.

2 Rules and records

Table 1 gives each sport's cap and the records we read. The Electronic Companion, the paper's online appendix, lists the key terms and the panel behind each result. We write n_f for the players a team starts with, and $n_{\max}$ for the players a team may bring on. In a re-entry sport the rule limits interchange events rather than players, and we write that event limit n_e. Some matches allow extra replacements, so we also use the *effective limit* n_{eff}, the limit that actually applies in that match.

Every record comes from public match data. Football's events and appearances come from a public dataset built from Transfermarkt's match pages (Cariboo, 2026). The other sports' records come from the leagues' own pages and statistics interfaces (MLB Advanced Media, 2026; National Hockey League, 2026; Australian Football League, 2026; AFL Tables, 2026; National Rugby League, 2026; It's Rugby, 2026; Euroleague Basketball, 2026; Gaelic Athletic Association, 2026). We call the records read for one analysis its *panel*. We read each panel one team-match at a time, and a *team-match* is one team's record of one match. The three-substitution panels leave out the matches played under the temporary five-substitution amendment of 8 May to 31 July 2020. The count and timing panels leave out the 2019–20 season altogether. Its matches under the amendment are kept apart

Table 1: The rules and the records. The limit is $n_{\max}$ in the one-way sports and the event limit n_e in rugby league and Australian rules. 'Not counted' lists the replacements the rule leaves outside the limit. The Electronic Companion gives each rule with its reference and explains season names.

Sport	n_f	Limit	Not counted	Seasons	Panel and source	Team-matches
Association football	11	3; 5 from 2020 by the competition's choice, in at most three stoppages in play; +1 in extra time	concussion from 2020	2012–25	29 men's leagues, Transfermarkt; event records name the player replaced in nine	59,982 (five); 58,328 (three); 42,303 (identity)
women				2024	seven leagues, Soccerdonna and FBref; the month-by-month identity only	1,996
Rugby union	15	7, then 8 by league from 2008; 8 in the Laws from 2014; up to four returns in the French leagues, 2018–25	blood, head injury	2004–26	Five leagues, itsrugby.fr; the identity also on the 2025–26 Top 14 sheets	25,028 (identity); 25,852 (count)
Gaelic football	15	5 to 2013, 6 in 2014–19, 5 from 2020; +3 in extra time	blood, head injury	2002–24	All-Ireland senior finals, match reports	10
Hurling	15	5; +3 in extra time	blood, head injury	2003–25	All-Ireland senior finals, match reports	18
Baseball	10	none; at most 13 pitchers from 2022; a pitcher faces three batters from 2020	–	2008–25	MLB Stats API game logs; the identity on 2024	77,096 (count); 4,858 (2024)
Rugby league	13	10 to 2015, 8 from 2016; bench 4	head injury, 18th player	2013–25	NRL match-centre feed	5,061
Australian rules	18	none to 2013, 120 in 2014–15, 90 in 2016–20, 75 from 2021; bench 4, 5	blood, head injury, changes at the breaks, the substitute	2013–25	AFL Tables and the AFL's statistics API, men and women	2,916 (men); 198 (women)
Ice hockey	6	none; 20 dressed	–	2008–24	NHL game sheets and shift charts, public statistics API	2,080 (2023); 730,501 skater-games
Basketball	5	none; 12 dressed	–	2007–24	EuroLeague and EuroCup box scores, 29 competition-seasons	10,830

to test the model of Section 5, and we call them the *held-out matches*.

In the re-entry sports, the minutes t_i of a team's players should add up to the players on the field times the length of the match t_M, so that $\sum_i t_i = n_f\, t_M$. Injury researchers call this total *exposure* (Fuller et al., 2006). Each of 10 panels in two sports passes this check on average. In the Australian Football League (AFL), the top competition of Australian rules, the sum is within 0.03 per cent of that total.

3 How many starters leave

3.1 Who plays the whole match

We say that a starter plays the whole match when the starter is on the field from the start to the final whistle. Under small caps, about half or more of the starters do. In football, 73 per cent of the starters played the whole match under three substitutions and 60 per cent under five. In rugby union, with a bench of eight, about half the starters did. In the All-Ireland finals of Gaelic football and hurling, 75 and 71 per cent did. In the National Rugby League (NRL), the top rugby league competition of Australia and New Zealand, a player may come back on. Yet 63 per cent of its starters played the whole match under eight interchanges, more than in football under five substitutions. Under a cap several times the size of the team, as in Australian rules, almost everyone comes off. Only 1 to 3 per cent of its appearances last the whole match.

In football and rugby league, position matters more than the cap. In rugby league the backs lasted the whole match in 91 per cent of their appearances under eight interchanges. The forwards, whom the interchanges rotate, lasted it in 31 per cent. The cut from ten interchanges to eight moved neither group by more than 1.1 percentage points. In football the goalkeeper finished 99 per cent of matches under

both caps. An outfield starter went from 71 to 56 per cent. Position also matters where there is no cap. Baseball's starting batters stay to the end in 84 to 88 per cent of games, but its starting pitchers in only 0.6 per cent. Where players change on the fly, as in ice hockey and basketball, almost no one plays the whole match.

3.2 The counting behind it

Counting explains why so many starters finish where no one can return. In a one-way sport each substitution takes one player off, and that player cannot come back unless the change was temporary. The player taken off is either a starter or a substitute who came on earlier. We call the second kind of substitution a *chain*. So every substitution that is not a chain takes off one starter. A starter recorded as leaving before the end of the match is *withdrawn*, and a team's *withdrawal rate* s_w is the share of its starters withdrawn.

Proposition 1. *In a one-way sport, let a team-match have S substitutions, of which C are chains, L take off a starter whom the record credits with the full match, and X take off a player on neither the starters' nor the substitutes' list. Let D be the starters sent off and not replaced, and U the other starters recorded short with no substitution or card. Then*

$$n_f \left(\frac{S}{n_f} - s_w \right) \;=\; C + L + X - D - U, \tag{1}$$

and, because $S \le n_{\text{eff}}$,

$$s_w \;\le\; \frac{n_{\text{eff}} + D + U}{n_f}. \tag{2}$$

The proof is only counting, and the Electronic Companion writes it out. In words, a team's substitution rate S/n_f, its substitutions per

player on the field, equals its withdrawal rate apart from five corrections. Of the five, only chains are a coach's choice. A *late entry* is a change made at minute 90 or later, in the added time at the end of the match. The record lists it at minute 90, so its starter seems to finish. Late entries and players on neither list come from the record, not the match. Starters sent off and other exits without a replacement are *departures without replacement*. They are events of the match, and the record gets them right. Take a football team that makes five substitutions. One of them takes off a player who had come on earlier, so it is a chain. Another comes in added time, and the record lists its starter as finishing the match. The record then shows three of its 11 starters withdrawn, not five. Each of the two missing starters is one of the five corrections. The second equation is the bound the cap sets. Its substitution part, n_{eff}/n_f, needs only the rulebook. A football team allowed five substitutions can take off by substitution at most five of its eleven starters, 45.5 per cent, whatever its coach decides.

3.3 Almost every change takes off a starter

Football's event records name, for every substitution, the player who left and the player who came on. So in nine leagues every part of equation (1) can be counted, on 42,303 team-matches. We call the share of substitutions that are chains the *chain share* $s_c = C/S$. In these leagues it is only 0.2 to 0.9 per cent. We call one league under one substitution rule a *league-regime*. Figure 1 places every league-regime with the substitution rate across and the withdrawal rate up. We call the difference between the two rates the *gap*, and give it in per cent of the team. Every one of football's 18 league-regimes has a gap below 3 per cent, which is 0.33 of a starter per team-match. We call this level the *tolerance*. Rugby union's lineups, which give both counts from one list, show a gap of 0.2 per cent. The gap also stays small in each month

of the season. Its largest monthly value in football's European leagues is 2.3 per cent, inside the tolerance.

3.4 Checking the records

On records that name the player replaced, every part of the identity can be counted. We flag a team's record when it has at least one late entry or one player on neither list. A flag needs only the record's list of changes and its players' minutes. In the records here 8 to 27 per cent of the teams' records are flagged. Every flag is a late entry, because the source lists added time at minute 90. Per team-match the flags rose in every league when teams could make five substitutions (Figure 2a). Counted on the records, the gap is mostly these late entries: chains are a sliver, and departures without replacement pull it the other way (Figure 2b).

4 How many changes a team makes

The number of changes varies little. A common measure of spread is the standard deviation, the typical distance from the average. For the number of changes a team makes, it is only 0.44 to 1.25 changes. Most teams use the whole cap: 85 per cent did under three substitutions and 56 per cent under five. The standard model of random match events is a Poisson count (Maher, 1982; Dixon & Coles, 1997). It spreads teams far too widely, and in football it puts 11.1 per cent of teams in the wrong count (Figure 3).

A narrow curve, cut off at the limit. We summarize the count with a normal distribution, the bell-shaped curve, with centre μ and standard deviation σ. We round it to whole changes and cut it off at the limit, and we call this curve the *censored normal*. For each sport

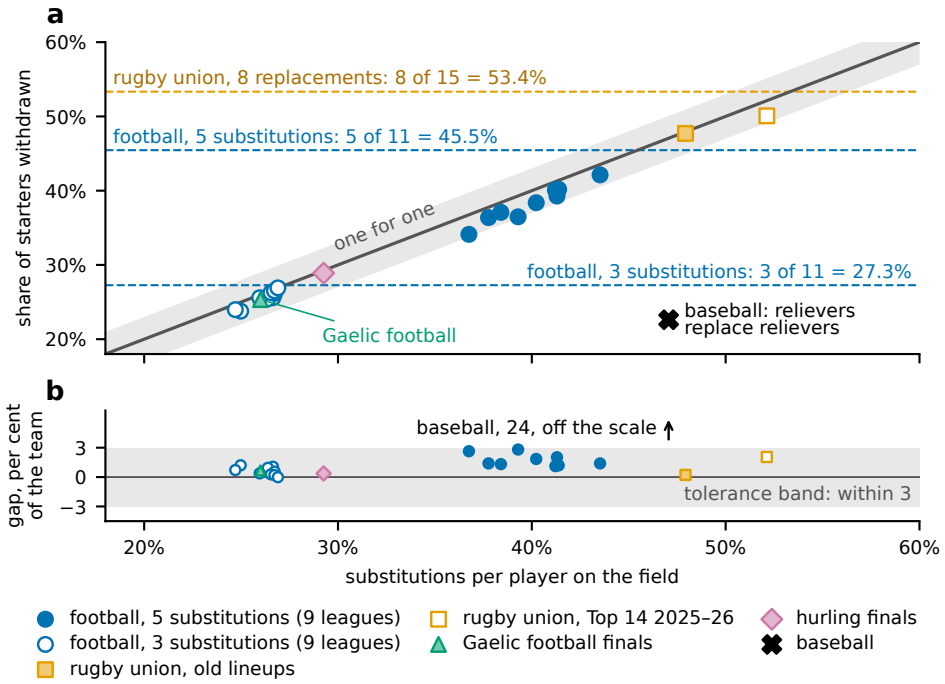


Figure 1: Substitutions take off starters almost one for one. (a) Each point is one league under one substitution rule, a league-regime, with its substitutions per player on the field across and the share of its starters withdrawn up. The diagonal is one starter withdrawn per substitution, and the grey tolerance band spans 3 per cent of the team either side of it. Football's filled points are under five substitutions and its open points under three, and each dashed line is the ceiling a cap sets, its share rounded up. Rugby union's old lineups and the Gaelic finals give both counts from one list, so their agreement is close to built in (lighter markers). The open square is France's Top 14 in 2025–26, on the newest lineups, which list every player's minutes. Baseball lies far below the band, because at least 45 per cent of its substitutions take off a player who had come on earlier, a chain. (b) Each gap, the substitution rate less the withdrawal rate, in per cent of the team, with the same band. Figure EC1 draws each gap with its 95 per cent interval, which shows how precise it is.

Alt text: Panel a: a scatter with a diagonal line inside a grey band. The eighteen football points lie in the band below their dashed cap lines, with the rugby union, Gaelic football and hurling points, and one baseball point sits far below it. Panel b: the same points as gaps near zero inside a narrow band, with an arrow for baseball off the top.

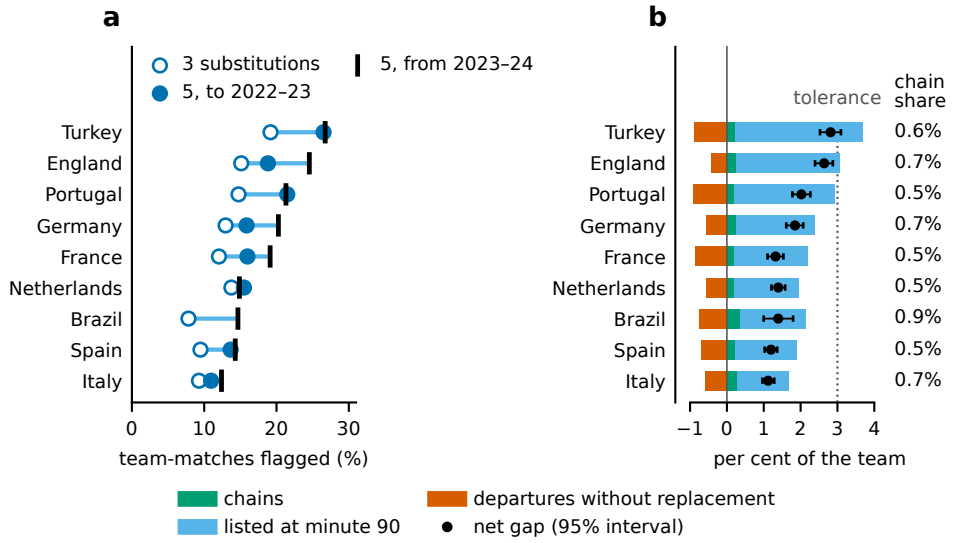


Figure 2: Counting flags 8 to 27 per cent of football's team-matches, every flag a change listed at minute 90. Per team-match the flags rose in every league at the move to five substitutions, and in England and Germany most of the rise came from 2023–24. In England that was the season the Premier League began to add the exact time lost. (a) The share of team-matches flagged in each league, pooled over its seasons under three substitutions (open), under five before 2023–24 (filled) and under five from 2023–24 (black tick), joined in that order; a team-match is flagged when its record credits a starter taken off by a substitution with the whole match. Brazil's only season under five substitutions is 2024, so it has a tick and no filled circle. The two rules' seasons are different years. (b) Football's gap under five substitutions, part by part, in per cent of the team and in the rows of (a): chains, the substitutions that take off a player who had come on earlier; changes listed at minute 90; and, to the left, departures without replacement, starters sent off or leaving unreplaced. By Proposition 1 the parts sum to the gap exactly. The black mark is the net gap with its 95 per cent interval, the dotted line its tolerance, and the right column the chain share, the chains' share of the substitutions.

Alt text: Panel a: for nine football leagues, a line runs from an open circle through a filled circle to a black tick; every filled circle lies right of its open circle, and the tick lies furthest right of the filled circle in England and Germany and close to it in the Netherlands, Portugal and Turkey. Panel b: for the same leagues, horizontal bars made mostly of changes listed at minute 90, with a thin green sliver of chains, a shorter bar to the left for departures without replacement, a black dot for each net gap and a dotted line for the tolerance to the right of every dot.

we fit it, choosing the centre and width that match the data best. Its centre and width describe the curve, not any one team's plan. It puts only 0.6 per cent of football's teams in the wrong count, and at most 4.2 per cent in the other count panels. Its width, its standard deviation before the cap cuts it, is 0.95 to 1.76 changes in every count panel. Its centre moves with the limit. In rugby union the centre is 2.20 replacements higher in the seasons with a bench of eight than in the earlier seasons with seven. It can move by more than one replacement, because it is the centre of the curve before the cut.

5 When the changes come

A simple line for the timing. Consider the teams that make exactly n changes in a match. Take the k-th of those changes and average its minute across the teams. We call that average a *cell*. Now picture n moments dropped at random on a stretch of the match and put in order, which statisticians call order statistics. On average the k-th of them falls at the fraction $k/(n+1)$ of the stretch, a classical result of probability (Cunnane, 1978; David & Nagaraja, 2003). A team that makes three changes, for example, would put the first a quarter of the way through the stretch, the second halfway and the third three quarters of the way. We call the stretch of the match in which a team's changes fall its *window*. If a team's changes behave like such random moments on a window, the cells fall on a line,

$$\mathrm{E}[t_{k:n}] \;=\; t_M \left(\zeta_{0,n} + \zeta_1 \, \frac{k}{n+1} \right), \tag{3}$$

where $\zeta_{0,n}$ is where the window opens for a team with n changes and ζ_1 how wide it is, both as shares of the match. We call this the order-statistic line.

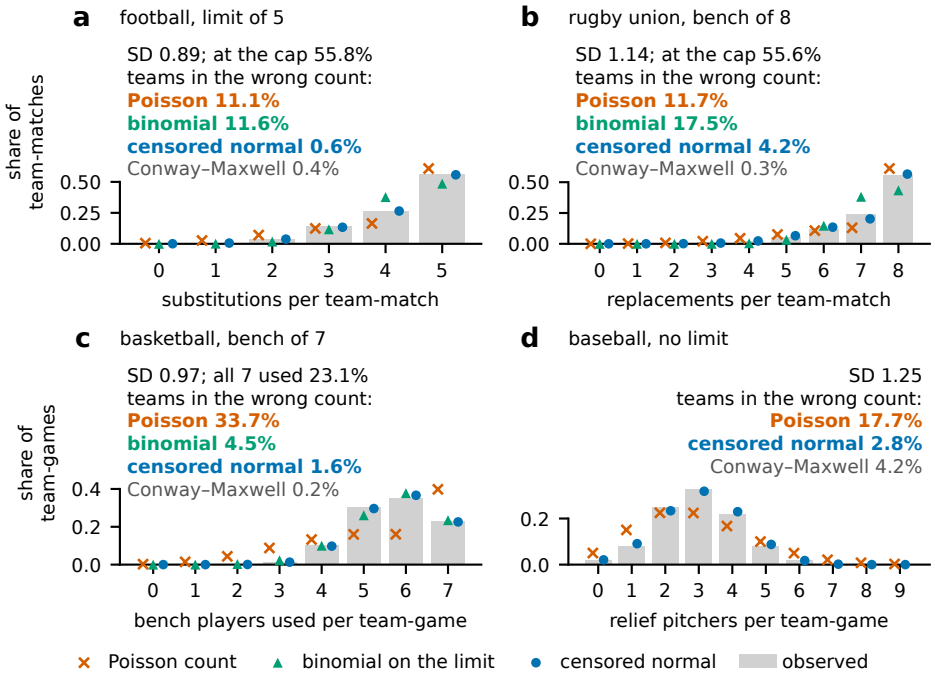


Figure 3: The number of changes varies little: the censored normal comes close to its shares, and a Poisson count does not. Grey bars are the observed shares, crosses the best-fitting Poisson count, triangles the binomial count on the limit with the observed mean (none in d), and dots the censored normal, each cut off at the limit. Each panel prints the observed standard deviation (SD), the share of teams at the cap (in basketball, the share that use all seven bench players), and the share of teams each model puts in the wrong count, that is, the share it would have to move to match the observed shares; its last line is the best Conway–Maxwell family, a standard model for counts narrower than Poisson, which is not drawn and which Figure EC3 compares. The Poisson count spreads teams too widely, and in football and rugby union the binomial is too narrow, because many teams use the whole cap and some use only a few changes. (a) Football under five substitutions. (b) Rugby union's sides with a bench of eight. (c) Basketball's bench players used, of the seven on its bench. (d) Baseball's relief pitchers, where no limit binds, shown to nine.

Alt text: Four bar charts of the number of changes per team. Blue dots sit on top of the grey bars. Red crosses miss them, too high at low counts and at the limit. Green triangles pile too much weight one change short of the limit in football and rugby union.

The line summarizes the averages. Figure 4a shows every cell of football's five-substitution era and of rugby union, and the line passes close to almost every cell. It misses the cells' average minutes by 1.1 minutes in football and 1.3 in rugby union. Putting every change at the average minute would miss by 8.7 and 7.8 (Figure 4c). With an opening for each number of changes, the line shows that teams that make more changes start earlier. Their window opens at 51.3 minutes for three changes, 50.8 for four and 48.5 for five.

Stoppages, and the bundling they require. The line gives the average minute of each change, but a simulator needs the changes of whole matches. Minutes picked at random, one by one, put two consecutive changes in one minute 6 per cent of the time. In the records 37 per cent share a minute. The reason is that a team using five substitutes may make them in at most three pauses during play. We call each pause in which a team makes one or more changes a *stoppage*.

Proposition 2. *Let a team stop play for changes at most w times during play, besides half-time, and let it make n changes, h of them at half-time. Then at least* $\max(0,\, h-1) + \max(0,\, n-h-w)$ *of its* $n-1$ *consecutive pairs of changes share a stoppage.*

Half-time holds its h changes in one stoppage, and the $n-h$ changes made during play fall in at most w stoppages. So the fewer stoppages a team may use, the more of its changes must share one. With w at three and each team's own h, football's rule forces 35.0 per cent of consecutive changes into one stoppage. So the rule forces most of the bunching in the records.

A model of the stoppages. We built a model of the stoppages in three steps from football's five-substitution seasons, which we call the

fitted seasons. A stoppage here is the changes listed in one minute. First, a team's number of changes is picked at random from the censored normal of Section 4. Second, the team splits its changes between half-time and its stoppages in play, in the proportions that teams with the same number of changes used. Third, a team with m stoppages in play picks their minutes at random from the minutes at which teams with m stoppages in play stopped play. It picks again if two minutes are the same, and puts them in order. So its j-th stoppage in play comes at the j-th of m ordered random minutes,

$$\begin{aligned} &\tau_{j:m}, \qquad \tau_1, \ldots, \tau_m \text{ drawn independently from } F_m, \\ &\text{a set with a repeated minute being redrawn,} \end{aligned} \tag{4}$$

where F_m is the distribution of the minutes at which teams with m stoppages in play stopped play. Each change then takes its stoppage's minute. The order-statistic line is the case in which every change is its own stoppage and F_m is spread evenly over the window. On the fitted seasons the model misses the cells' average minutes by 0.6 minutes, against 1.1 for the line.

We then tested the model on the held-out matches, which no part of this paper had read. They are the 1,270 team-matches of the 2019–20 season played under the temporary amendment of May 2020. We wrote down six predictions with their pass marks in a register before reading them, and every one held (Electronic Companion). The model missed the cells' average minutes by 0.9 minutes, against 1.5 for the line. It put 37 per cent of consecutive changes in one minute, against 33 per cent in these matches' records (Figure 4b).

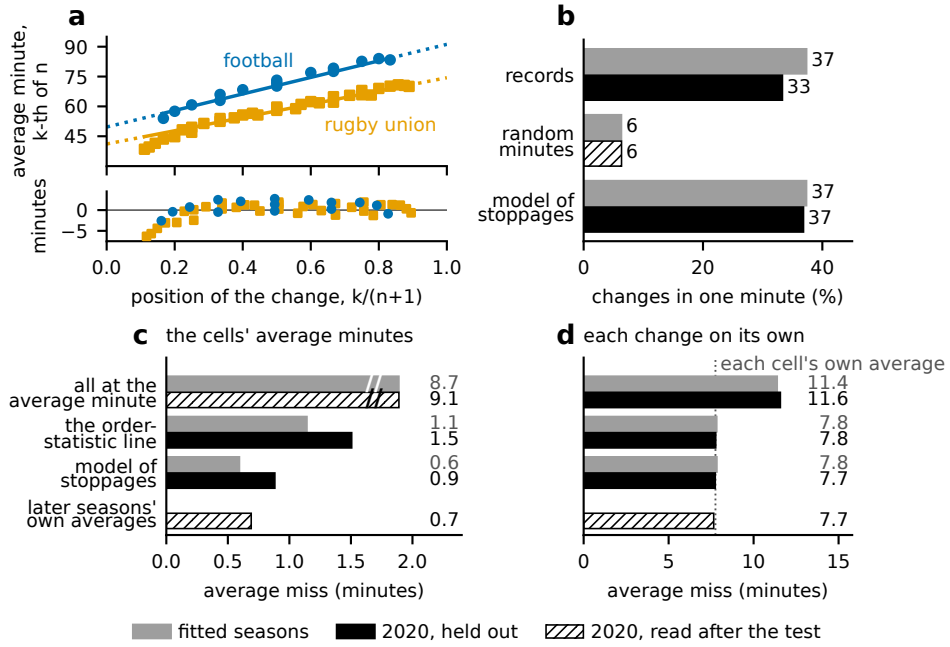


Figure 4: The average minute of each change can be read from the number of changes, and changes come bundled in stoppages. (a) Each point, a cell, is the average minute of the k-th of n changes in football's five-substitution era and in rugby union, with the order-statistic line, drawn here with one opening shared by all n, dotted past the cells; the strip gives each cell's distance from it in minutes. First changes come early, partly for injuries. Rugby union's cells sit lower because its match lasts 80 minutes. (b) Consecutive changes made in one minute: in the records, for minutes picked at random and for the model of stoppages, which is built to match the fitted seasons. (c) How far each placement misses football's cell averages, each cell weighted by its changes, on the fitted seasons' 15 cells and the 14 held-out cells with at least 50 changes; the average minute's bars are cut. The later seasons' own averages, with no grey bar as the fitted seasons are their source, miss slightly less than the model, a difference whose 95 per cent interval includes zero. (d) The same for each change on its own; the dotted line is the miss of each cell's own average on the fitted seasons, 7.8 minutes. In (b) and (c) the black bars answer predictions registered before the held-out matches were read; hatched bars were read after the test.

Alt text: Panel a: points along two rising lines, football above rugby union, most within two minutes of their line. Panels b to d: bars, short in b for random minutes, cut in c for the average minute, near eight minutes in d.

6 Discussion

6.1 Managerial implications

For governing bodies. The counting gives governing bodies a simple formula. A limit of n_{eff} caps the share of starters a team can withdraw by substitution at n_{eff}/n_f. So it guarantees that at least $1 - n_{\text{eff}}/n_f$ of them finish, less departures without replacement. We call this share the *floor*. Figure 5a turns this formula into a design chart, and Figure 5b sets the spread between positions beside each cap. Teams came close to the floor under the caps they had. The share of football's starters who finished sat 0.4 percentage points above it under three substitutions and 5.3 under five. Rugby union's sat 3.3 to 5.6 points above it. How close teams would come to a new cap is a forecast. A sixth substitution, for example, would raise football's ceiling from 45.5 to 54.6 per cent of the team. At least 45.4 per cent of the starters would still finish. The Gaelic Athletic Association's football advisory group has proposed a sixth substitute (Harrington, 2026). It would lower Gaelic football's floor from two thirds of the starters to 60 per cent. Proposition 2 does the same for stoppages. With today's numbers of changes, two stoppages in play would force at least 62 per cent of consecutive changes into one stoppage, and four at least 14 per cent. Every rulebook here that sets a limit now allows replacements for head injuries outside it. So the formula also shows how many more starters each exemption lets a team take off. The formula is one input among several, beside player welfare, tactics, fairness, the flow of the match, spectators, heat and cost (Electronic Companion). It does not rank these goals, but it tells a governing body exactly how far each option moves the share of starters who can be withdrawn.

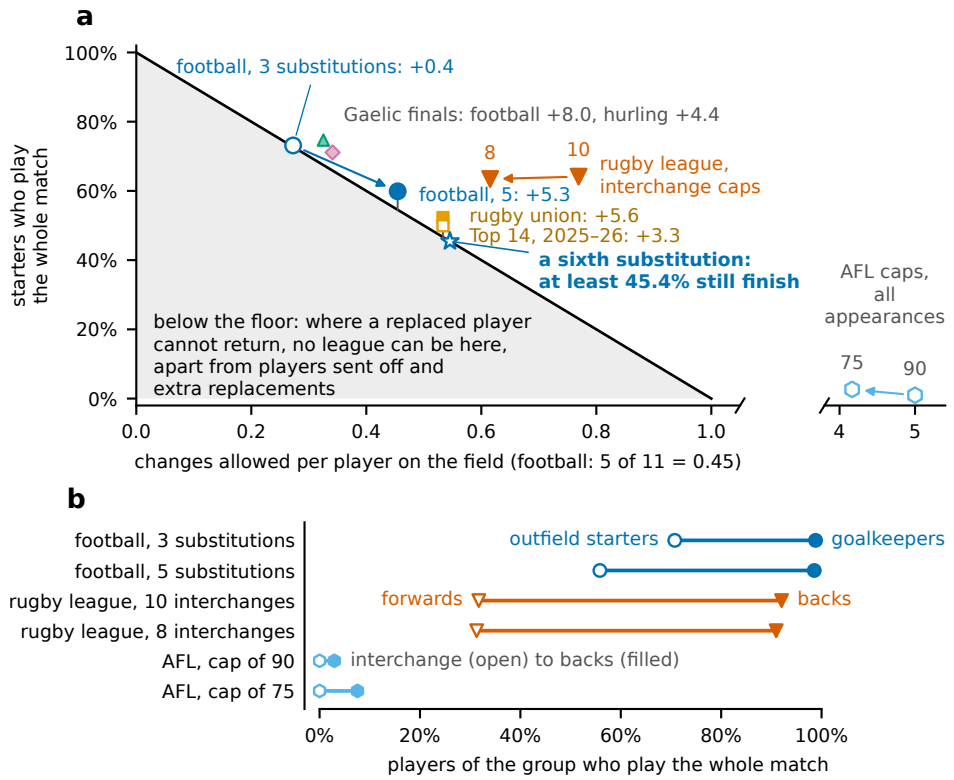


Figure 5: A cap sets a floor under the starters who finish, and the leagues where a replaced player cannot return sit close above it. (a) Where a replaced player cannot return, at least $1 - n_{\text{eff}}/n_f$ of the starters finish, less those sent off: the line is this floor, and the region below it is shaded. Labels and thin grey lines give each league's distance above it in percentage points; the Gaelic finals are plotted at five substitutes, though the 2015 football final allowed six. Football is read at the final whistle, and rugby union's open square is the Top 14 on its newest lineups. Arrows join each change of cap. Rugby league and the AFL let players return, so their caps count interchanges; the AFL sits in a compressed segment. (b) Under each cap, the group of positions that plays the whole match most often (filled) and least often (open). In football and rugby league the groups differ more than a change of cap moved them, and in the AFL almost no one plays the whole match.

Alt text: Panel a: a falling floor line over a shaded region; football's and rugby union's points sit just above it, the lighter Gaelic points a little higher, a star on the floor, rugby league's points higher to the right and the AFL's near zero. Panel b: six bars between two markers, long for football and rugby league, short near zero for the AFL.

For statistics units and data providers. In sports where no player may return even for a time, the identity gives a check at two levels. On records that name the player replaced, a data provider can flag each team-match with a late entry or a player on neither list. Where records do not pair players, as in most public appearance tables, the gap checks them across many matches once the starters sent off are added back. A sum near zero then points to changes missing from the record, as in a first reading of Brazil's top league (Electronic Companion). Such a source's minutes also stop at minute 90 for every player, so injury research must add the added time back to measure exposure. A provider that keeps added time in its minutes, as Wyscout does (Wyscout, 2026), raises no flag.

For builders of match simulators. A simulator can pick the number of changes at random from a censored normal. It can then pick the team's stoppages as equation (4) describes, and give each change its stoppage's minute. On the held-out matches, the censored normal gave 4.9 per cent of teams the wrong number of changes, against 13 per cent for a Poisson count. The model put 37 per cent of consecutive changes in one minute, against 6 per cent for minutes picked at random and 33 per cent in the held-out records. For each change on its own, the model and the line miss alike (Figure 4d), so the model's gain is in how it groups changes. The Electronic Companion's tables let a simulator run the model.

What the paper does not do. The paper makes no forecast of how teams would respond to a new limit and draws no link to results, because both need study designs this paper does not have.

6.2 Limitations

Eight limitations apply. First, the agreement between the two rates has a boundary. Baseball is strictly one-way, yet its gap is 24 per cent, because each relief pitcher after the first replaces a pitcher who had come on as a substitute. So a small gap needs few chains as well as a ban on re-entry.

Second, the tolerance of 3 per cent was set on football after football's gaps had been read, so the gaps are descriptions, not tests. The predictions for the other sports were entered in the register only after their tests had been written.

Third, rugby union's gap includes temporary replacements, the sin-bin and the front-row replacement rules. The Gaelic panels hold only the finals with complete and consistent lineups. The AFL's own interchange records may make its exposure check pass automatically.

Fourth, of the 39 predictions in the paper's main register, nine failed and three could not be checked (Electronic Companion).

Fifth, the count models are summaries rather than laws. In rugby union and basketball a Conway–Maxwell family, a standard model for counts narrower than Poisson, fits better than the censored normal (Shmueli et al., 2005; Kadane, 2016).

Sixth, the window describes the middle of the sequence, not its ends, and in rugby union even the middle changes reject one window. The model of stoppages was tested on one short period under the same rule, and its pass marks were set after the fitted seasons had been read. After the test, the later fitted seasons' own cell averages missed slightly less than the model, and copying whole sequences from real matches did better still. So four predictions are registered for the 2026–27 seasons, the first under the ten-second exit, whose records we have not read.

Seventh, the comparisons across rules are descriptions, not effects,

because football's three- and five-substitution seasons are different years.

Last, the analyses use different samples, and limits are themselves outcomes of rule-making (Australian Football League, 2013).

7 Conclusion

The substitution limit works as a speed limit does. It caps how many starters a team can take off, and the coach decides whom to keep on. In football and rugby league, who plays the whole match depends more on position than on the cap. Where a replaced player cannot return, one line of arithmetic ties the starters withdrawn to the substitutions made. It gives governing bodies a formula and data providers a free check. The arithmetic does not say how teams will respond to a new limit. The number of changes and their timing follow regularities that simulators can use. The model of stoppages met its registered predictions on matches it had not seen.

Data availability

The deposits behind every number in this paper, the software that produces them, and a ledger that maps each printed number to its deposit will be archived with the paper in a public repository. No login, licence or fee was involved, and the deposits hold only aggregates, never a named player. The fits use SciPy (Virtanen et al., 2020) and the figures Matplotlib (Hunter, 2007). The archive identifier is withheld for anonymized review.

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